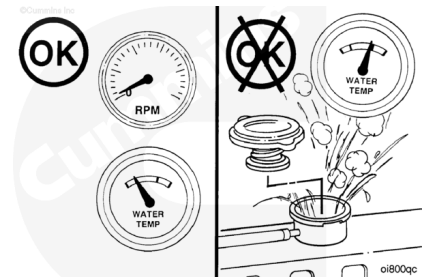


Trainings ▾

General Information

⚠ WARNING ⚠

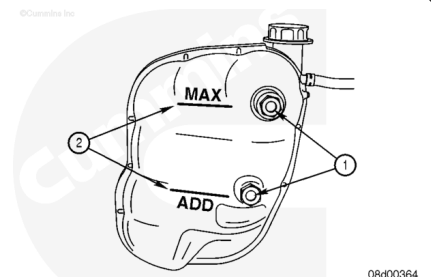
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



The engine coolant level **must** be checked daily.

⚠ CAUTION ⚠

Do not add cold coolant to a hot engine. Engine castings can be damaged. Allow the engine to cool below 50°C [120°F] before adding coolant.



⚠ CAUTION ⚠

Do not use the coolant replacement tool to evacuate contaminated coolant or a system that is suspected of contamination. This could result in cross contamination of coolant. The coolant drain method should be used for removing contaminated coolant. Refer to the equipment manufacturer's service manual for specific instructions on cleaning the tool of contaminants.

Note : When the vehicle/equipment or engine is equipped with a quick disconnect fitting in the cooling system package, the Coolant Replacer Method is the preferred method for coolant removal. Use the coolant replacer tool, Part Number 2892459. If the vehicle/equipment or engine does not have a quick disconnect fitting presently installed, one can be installed in the cooling system package to utilize the Coolant Replacer Method; otherwise the Coolant Drain and Fill Method must be used.

Note : In the event of a water pump or exhaust gas recirculation (EGR) cooler malfunction, check the coolant level switch or sensor for proper operation. Refer to the equipment manufacturer service manual for operational checks and repairs.

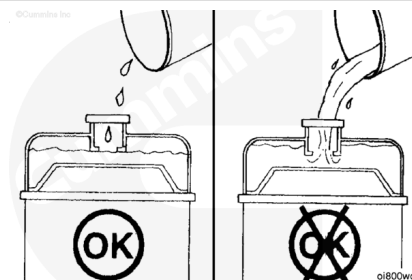
On applications that use a coolant recovery system, check to make sure the coolant is at the appropriate level in the coolant recovery tank for the engine temperature.

Many coolant recovery/expansion tanks, also called "top tanks", have sight glasses, or are made of a clear material (**not** shown) to aid in checking the coolant level without removing the radiator cap.

It is important to understand the impact of temperature on the expansion of the coolant. Most "top tanks" do **not** have a provision for a "FULL HOT" coolant level. Filling the "top tank" while hot will result in a low operating level once the system has cooled.

Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill or recovery/expansion tank.

Note : Some radiators have two fill necks. Both fill necks **must** be filled when the cooling system is drained.

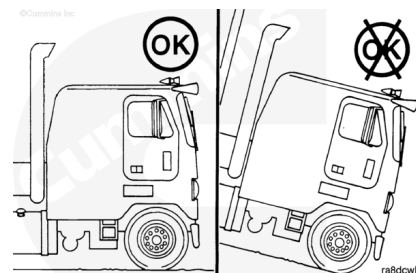


Coolant Replacer Method

Evacuation

The following steps are used to evacuate the cooling system using the coolant replacer tool, Part Number 2892459.

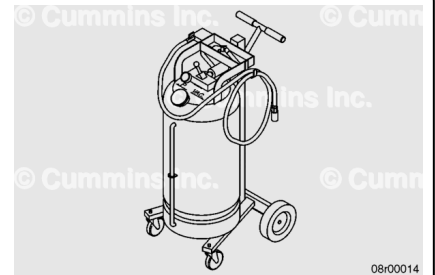
Note : When the vehicle/equipment or engine is equipped with a quick disconnect fitting in the cooling system package, the Coolant Replacer Method is the preferred method for coolant removal. Use the coolant replacer tool, Part Number 2892459. If the vehicle/equipment or engine does **not** have a quick disconnect fitting presently installed, one can be installed in the cooling system package to utilize the Coolant Replacer Method; otherwise the Coolant Drain and Fill Method **must** be used.



Position the equipment on level ground.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ CAUTION ⚠

Do not use the coolant replacement tool to evacuate contaminated coolant or a system that is suspected of contamination. This could result in cross contamination of coolant. The coolant drain method should be used for removing contaminated coolant. Refer to the equipment manufacturer's service manual for specific instructions on cleaning the tool of contaminants.

Note : See equipment manufacturer service information for special coolant drain requirements. Special instructions may also be located near the cooling system access point or fill door on the vehicle.

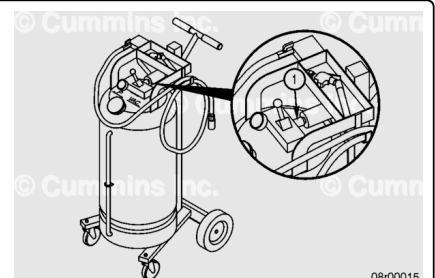
Isolate the engine from the vehicle cooling system by closing coolant flow valves to the equipment heating systems before starting the repair. This will prevent the heater circuit from draining, minimizing the chance for air pockets to be present during the fill process.

This air can be very difficult to purge in some applications with several feet of plumbing and multiple heater cores.

Remove the radiator cap.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



The coolant replacement tool tank capacity of 68 Liters [18 gal] is adequate for most applications. An additional storage tank can be used for cooling system packages with more than 68 Liters [18 gal] capacity.

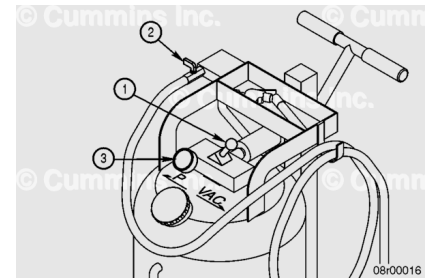
Be sure there is no air pressure in the coolant replacement tool tank by opening the pressure relief valve (1) located on the control block of the coolant replacement tool.

Connect the coolant replacement tool to a shop air supply regulated at 621 kPa [90 psi].

Switch the coolant replacement tool control lever (1) to "VAC" and leave the service hose valve (2) closed. This will create a vacuum in the tank to evacuate coolant from the cooling system package.

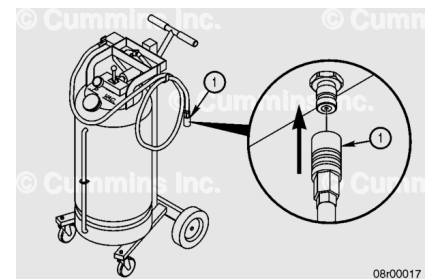
Monitor the gauge (3) and build a vacuum of approximately 508 mm-Hg [20 in-Hg] in the coolant replacement tool tank. Once the vacuum has been achieved, move the control valve lever (1) to the middle position.

Maintain approximately a 508 mm-Hg [20 in-Hg] vacuum to achieve a faster drain.



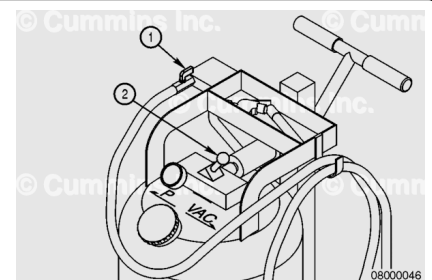
Attach the fill hose quick disconnect coupling (1) of the coolant replacement tool to the quick disconnect fitting. Location of the fitting may vary between original equipment manufacturers (OEMs), but the fitting is generally located in the lowest point of the vehicle/equipment cooling system package.

Note : Most Volvos are equipped with a different style fitting located in the radiator. An adapter hose is needed to connect the Cummins® coolant replacement tool to the fitting. The adapter hose is included in an accessory kit.



Open the service hose valve (1) by turning it **clockwise** until it is completely open (approximately ¼ turn).

Additional shop air may be required to maintain enough of a system vacuum to remove the coolant from the system. This can be done by moving the control valve lever (2) back to the VAC position.



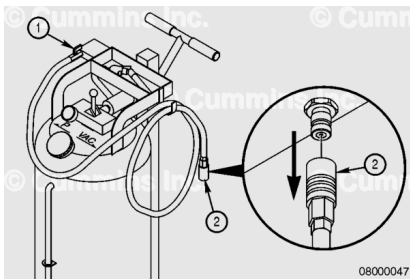
When the cooling system has been evacuated, a coolant and air mixture will be visible in the clear section of the coolant replacement tool fill hose.

Note : Some residual coolant will settle in the coolant package as the recessed areas of the block continue to

drain down over the next few minutes.

Once the system has been evacuated, turn the service hose valve (1) to the closed position by turning the valve **counterclockwise** a ¼ turn. Disconnect the shop air connection from the coolant replacer tool.

Disconnect the fill hose quick disconnect coupling (2) on the coolant replacement tool from the quick disconnect fitting.



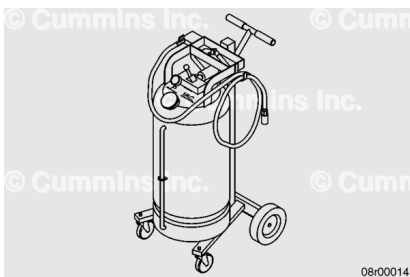
Injection

⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.

The following steps are used to inject coolant into the cooling system using the coolant replacer tool, Part Number 2892459.

Note : When the vehicle/equipment or engine is equipped with a quick disconnect fitting in the cooling system package, the Coolant Replacer Method is the preferred method for coolant removal. Use the coolant replacer tool, Part Number 2892459. If the vehicle/equipment or engine does not have a quick disconnect fitting presently installed, one can be installed in the cooling system package to utilize the Coolant Replacer Method; otherwise the Coolant Drain and Fill Method must be used.



Cummins Inc. recommends the use of either a 50/50 mixture of good quality water and fully-formulated antifreeze, or fully-formulated coolant when filling the cooling system. The fully-formulated antifreeze or coolant **must** meet TMC RP329 or TMC RP330 specifications.

Good quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.



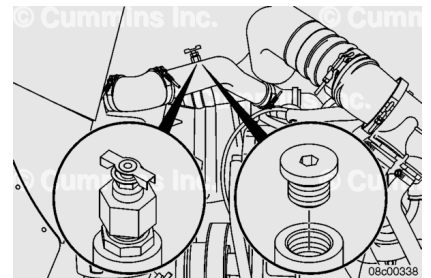
Water Quality	
Calcium Magnesium (Hardness)	Maximum 170 ppm as (CaCO ₃ + MgCO ₃)
Chloride	40 ppm as (Cl)
Sulfate	100 ppm as (SO ₄)

Cummins Inc. recommends the use of Fleetguard® ES COMPLEAT™. It is available in glycol forms (ethylene and propylene) and complies with TMC RP329 and RP330 standards.

Refer to Cummins® Coolant Requirement and Maintenance, Bulletin 3666132, ([/qs3/pubsys2/xml/en/bulletin/3666132.html](http://qs3/pubsys2/xml/en/bulletin/3666132.html)) for more engine coolant specifications.

⚠ CAUTION ⚠

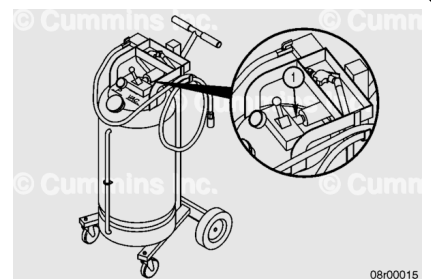
Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



Note : Some applications can have a manual bleed valve or pipe plug that is required to be opened to properly fill the system. The upper radiator pipe is a common location.

If applicable, open the manual bleed valve or remove the pipe plug before filling the cooling system.

Be sure there is no air pressure in the coolant replacement tool tank by opening the pressure relief valve (1) located on the control block of the coolant replacement tool.

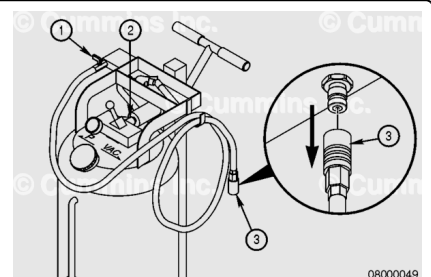


Connect the coolant replacement tool to a shop air supply regulated at 621 kPa [90 psi].

Attach the fill hose quick disconnect coupling (1) of the coolant replacement tool to the quick disconnect fitting located in the vehicle/equipment cooling system package.

With the service hose valve (2) in the closed position, switch the coolant replacement tool control lever (3) to “P” and build 172 kPa [25 psi] pressure on the gauge.

Slowly open the service hose valve (2) part way by turning it **counterclockwise** approximately 1/8 turn.



Do **not** open the ball valve completely as this will increase the coolant flow rate and increase the amount of air pockets created in the cooling system; This will provide a more complete injection of coolant.

Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill or recovery/expansion tank or until the coolant replacer tool is empty (whichever occurs first).

On applications that use a coolant recovery system, check to make sure the coolant is at the appropriate level in the coolant recovery tank for the engine temperature.

Many coolant recovery/expansion tanks, also called "top tanks", have sight glasses or are made of a clear material (**not** shown) to aid in checking the coolant level without removing the radiator cap.

It is important to understand the impact of temperature on the expansion of the coolant. Most "top tanks" do **not** have a provision for a "FULL HOT" coolant level. Filling the "top tank" while hot will result in a low operating level once the system has cooled.



⚠ CAUTION ⚠

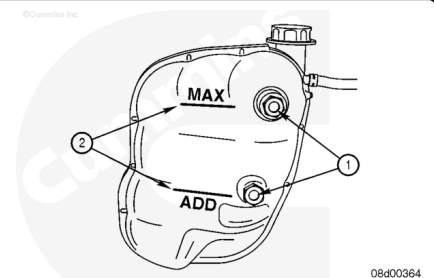
The cooling system must be filled properly to prevent air locks or serious engine damage can result.

Note : If all coolant drained from the system was collected, the same volume or more **must** go back into the system. If any drained coolant remains in the tool after filling, this is an indication of an air pocket in the cooling system package which **must** be purged before returning the vehicle to service.

Note : Top off of coolant might be necessary for repairs that were performed to correct a coolant loss issue.

If all coolant drained from the system would **not** return to the system, or the level is above the maximum level, this is an indication of an air pocket in the cooling system package, which **must** be purged before returning the vehicle to service.

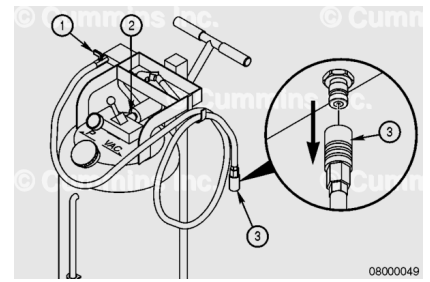
To remove an air pocket from the cooling system, the coolant replacement tool can be used. Refer to the Vacuum section of this procedure.



Once the coolant level has been returned to the correct level, close the service hose valve (1) by turning the valve **clockwise** until closed.

Remove pressure from the coolant replacement tool tank by opening the pressure release valve on the back of the control block (2).

Disconnect the service hose quick disconnect coupling (3) from the quick disconnect fitting of the vehicle/equipment.

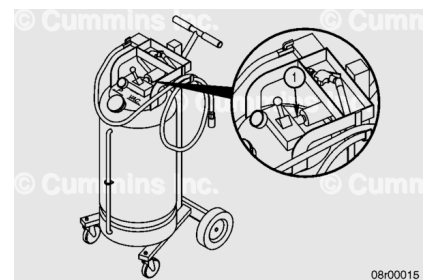


Vacuum

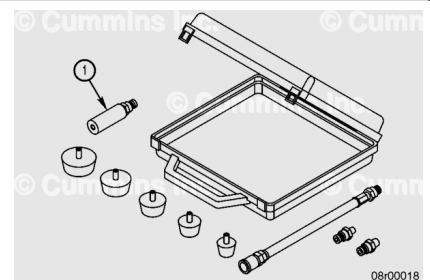
The following steps are used to place a vacuum on the cooling system using the coolant replacer tool, Part Number 2892459.

Be sure there is no air pressure in the coolant replacement tool tank by opening the pressure relief valve (1) located on the control block of the coolant replacement tool.

Clamp off any vent hoses/connections or overflow to the cooling system.



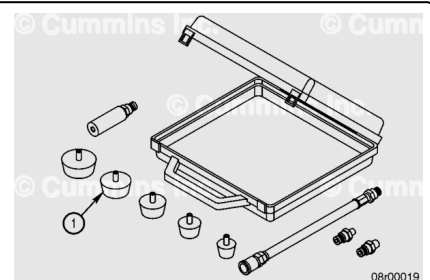
Attach the coolant dam handle (1) to the fill hose of the coolant replacement tool.



Attach the appropriate size coolant dam rubber adapter (1) onto the coolant dam handle.

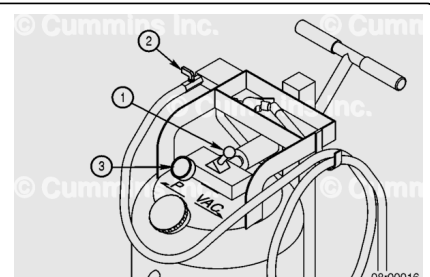
The size of the fill neck will differ between OEMs. Choose the appropriate sized coolant dam rubber adapters.

Connect the coolant replacement tool to shop air regulated at 621 kPa [90 psi].



Place the coolant dam over the coolant fill neck in the radiator or overflow tank.

The size of the fill neck will differ between OEMs. Choose the appropriate sized coolant dam rubber adapters.



Switch the coolant replacement tool control lever (1) to "VAC" and leave the service hose valve (2) closed. This will create a vacuum in the tank to evacuate coolant from the cooling system package.

Monitor the gauge (3) and build a vacuum of approximately 508 mm-Hg [20 in-Hg] in the coolant replacement tool tank.

Slowly open the service hose valve (2) by turning it **clockwise** until it is completely open, approximately $\frac{1}{4}$ turn.

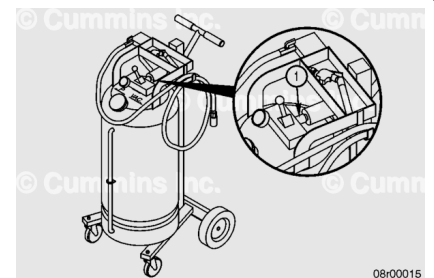
Once the cooling system is put into a vacuum, any air trapped in the cooling system will be evacuated through the top of the system. This is noticed as air bubbles through the overflow tank or top of the radiator.

When air bubbles are no longer being drawn to the top of the cooling system, move the control lever to the middle position. Remove the vacuum on the coolant replacement tool tank by opening the pressure relief valve located on the control block of the coolant replacement tool (1).

Remove the coolant dam from the radiator fill neck or overflow tank.

Disconnect the coolant replacement tool from the regulated shop air supply.

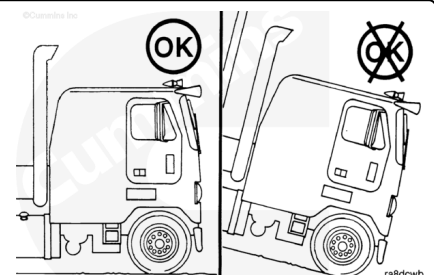
Install the radiator cap.



Drain

Note : This method is to be used when the vehicle/equipment or engine is **not** equipped with a quick disconnect fitting in the cooling system package. If the engine or vehicle/equipment is equipped with a quick disconnect fitting in the cooling system package, the Coolant Replacer Method must be used.

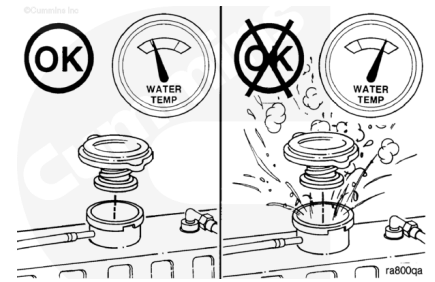
Position the equipment on level ground.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam

can cause personal injury.



Note : If the coolant is being changed, or if the cooling system is being flushed, leave the coolant flow valves to the equipment heating systems open, in order to completely drain the system.

Note : Refer to the equipment manufacturer service manual for special coolant drain requirements. Special instructions may also be located near the cooling system access point or fill door on the vehicle.

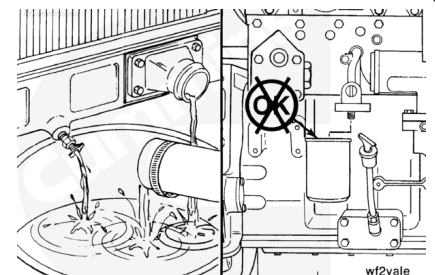
Isolate the engine from the vehicle cooling system by closing coolant flow valves to the equipment heating systems before starting the repair. This will prevent the heater circuit from draining, minimizing the chance for air pockets to be present during the fill process.

This air can be very difficult to purge in some applications that have several feet of plumbing and multiple heater cores.

Remove the radiator cap or expansion tank fill cap to allow the coolant to drain completely.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Note : Not all engines are assembled with a coolant filter. The use of coolant filter is optional when using coolant meeting Cummins® coolant requirements.

Do **not** allow the cooling system to dry out.

If the coolant is to be reused, the holding container **must** be free of oil and dirt.

A drain pan with a capacity of 57 liters [51 gal] is adequate for most applications.

Before adding coolant back into the engine, it **must** be tested for SCA (supplemental coolant additive) concentration.

For automotive applications, use the Signature™ and ISX Operation and Maintenance Manual, Bulletin 3666251. Refer to Procedure 018-004 in Section V.

(qs3/pubsys2/xml/en/procedures/00/00-018-004.html)

For industrial applications, use the Industrial and Power Generation QSX15 Engine Operation and Maintenance Manual, Bulletin 3666423. Refer to Procedure 018-004 in Section V. (</qs3/pubsys2/xml/en/procedures/101/101-018-004.html>)

Open the radiator drain valve(s).

Remove the lower radiator hose(s).

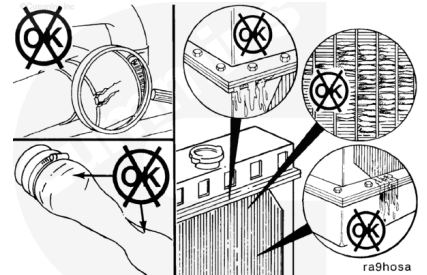
Do **not** remove the coolant filter.

Check for damaged hoses and loose or damaged hose clamps. Replace as required.

Check the radiator for leaks, damage, and buildup of dirt. Clean and replace, as needed.

Close the drain valve(s).

Install the lower radiator hose(s). See equipment manufacturer service information.



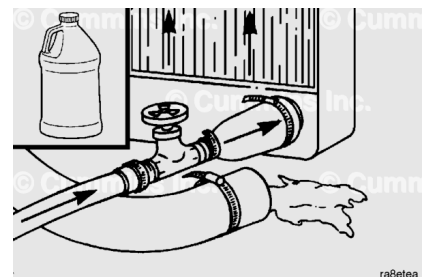
Flush

CM570

⚠ CAUTION ⚠

Do not use caustic cleaners in the cooling system. Aluminum components will be damaged.

The cooling system **must** be clean to work correctly and to eliminate buildup of harmful chemicals.



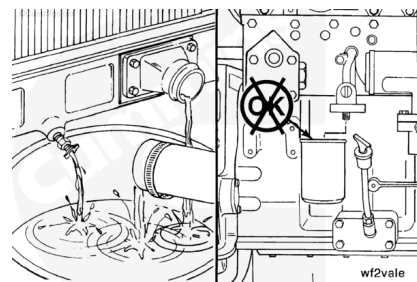
Restore™ is a heavy-duty cooling system cleaner that removes corrosion, silica gel, and other deposits. The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, longer cleaning times, or the use of Restore Plus™. Up to twice the recommended concentration levels of Restore™ can be used safely. Restore Plus™ **must** be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.



Drain the coolant completely.

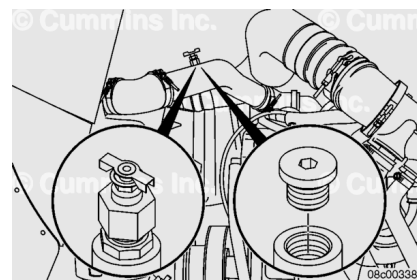
Close the radiator drain valve(s).

Install the lower radiator hose(s). Refer to the equipment manufacturer service information.



⚠ CAUTION ⚠

Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



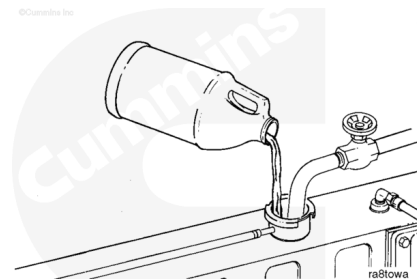
Note : Some applications can have a manual bleed valve or pipe plug that is required to be opened to properly fill the system. The upper radiator pipe is a common location.

Note : Some applications can have a pipe plug installed in place of the manual bleed valve.

If applicable, open the manual bleed valve or remove the pipe plug before filling the cooling system.

⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.



⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.

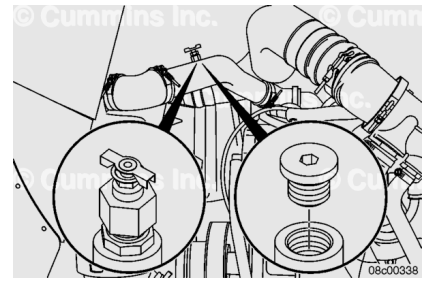
Note : The system has a designed fill rate of 19 liters [5 gal] per minute.

Add 3.8 liters [1 gal] of Fleetguard® Restore™, Restore Plus™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with clean water.

Turn the cab heater temperature switch to high to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

⚠ CAUTION ⚠

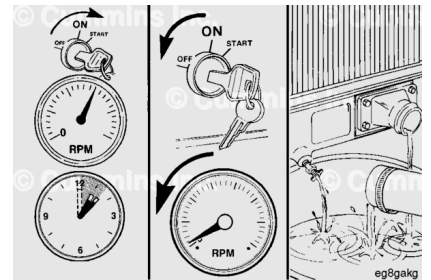
Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



If applicable, install the manual bleed valve or the pipe plug after filling the cooling system. Refer to the equipment manufacturer service information.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : All coolant shutoff valves should be returned to their open positions once the system has been refilled and the deaeration process is ready to begin. Refer to the equipment manufacturer service information for specific instructions.

Install the radiator cap or expansion tank fill cap.

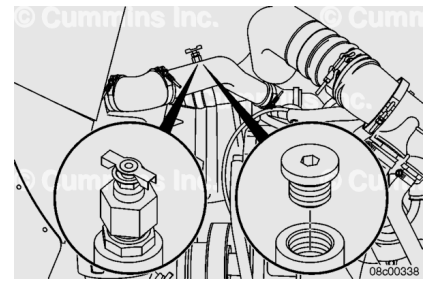
Operate the engine at HIGH IDLE and normal operating temperatures, at least 85°C [185°F], for 1 to 1½ hours.

Operate the engine at LOW IDLE for 2 minutes before shutting it down. This allows adequate cooling of pistons, cylinders, bearings, and turbocharger components.

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

⚠ CAUTION ⚠

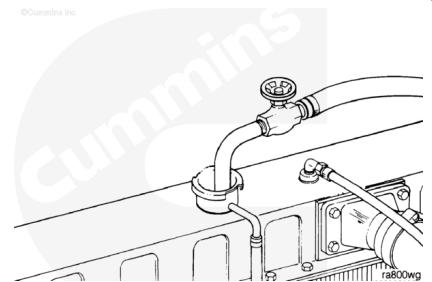
Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



If applicable, open the manual bleed valve or remove the pipe plug before filling the cooling system.

⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks, or serious engine damage can result.



Note : The system has a designed fill rate of 19 liters [5 gal] per minute.

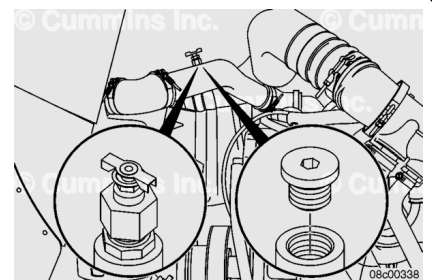
Close the radiator drain valve(s).

Install the lower radiator hose(s). Refer to the equipment manufacturer service information.

Fill the cooling system with clean water.

⚠ CAUTION ⚠

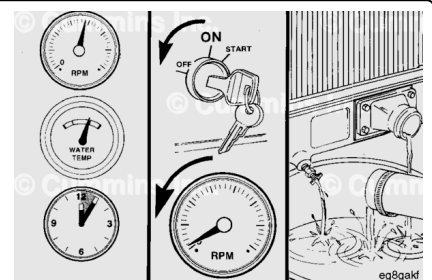
Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



If applicable, install the manual bleed valve or the pipe plug after filling the cooling system. Refer to the equipment manufacturer service information.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : All coolant shutoff valves should be returned to their open positions once the system has been refilled and the deaeration process is ready to begin. Refer to the equipment manufacturer service information for specific instructions.

Install the radiator cap or expansion tank fill cap.

Operate the engine at HIGH IDLE for 5 minutes with the coolant temperature above 85°C [185°F].

Operate the engine at LOW IDLE for 2 minutes before shutting it down. This allows adequate cooling of pistons, cylinders, bearings, and turbocharger components.

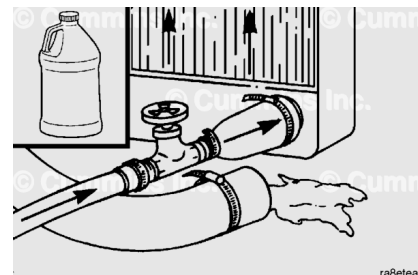
Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

If the water being drained is still dirty, the system **must** be flushed until the water is clean.

Automotive with CM870 and CM871

⚠ CAUTION ⚠

Do not use caustic cleaners in the cooling system. Aluminum components will be damaged.



The cooling system **must** be clean to work correctly and to eliminate buildup of harmful chemicals.

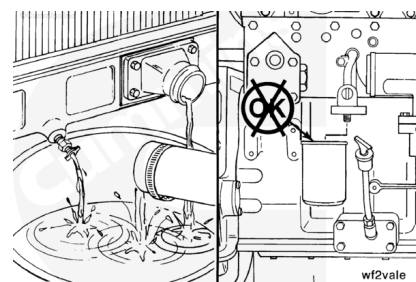
Restore™ is a heavy-duty cooling system cleaner that removes corrosion, silica gel, and other deposits. The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, longer cleaning times, or the use of Restore Plus™. Up to twice the recommended concentration levels of Restore™ can be used safely. Restore Plus™ **must** be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.



Drain the coolant completely.

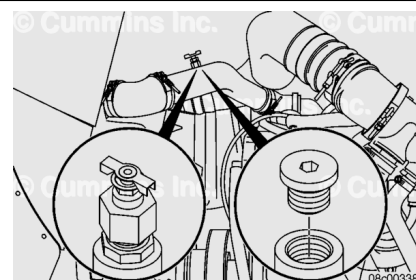
Close the radiator drain valve(s).

Install the lower radiator hose(s). Refer to the equipment manufacturer service information.



⚠ CAUTION ⚠

Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



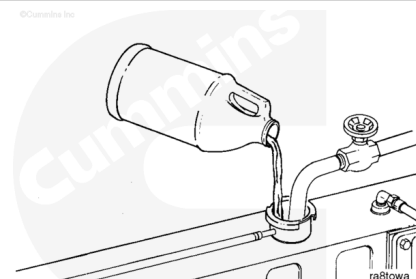
Note : Some applications can have a manual bleed valve or pipe plug that is required to be opened to properly fill the system. The upper radiator pipe is a common location.

Note : Some applications can have a pipe plug installed in place of the manual bleed valve.

If applicable, open the manual bleed valve or remove the pipe plug before filling the cooling system.

⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.



⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.

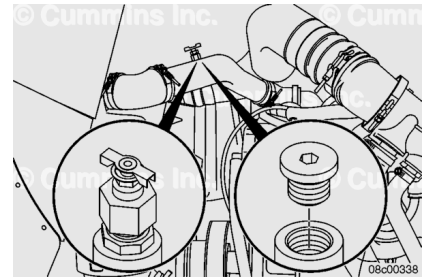
Note : The system has a designed fill rate of 19 liters [5 gal] per minute.

Add 3.8 liters [1 gal] of Fleetguard® Restore™, Restore Plus™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with clean water.

Turn the cab heater temperature switch to high to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

⚠ CAUTION ⚠

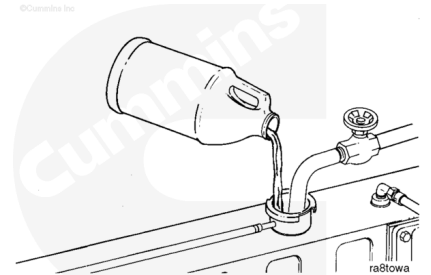
Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



If applicable, install the manual bleed valve or the pipe plug after filling the cooling system. Refer to the equipment manufacturer service information.

⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.



⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.

Note : Do not install the radiator cap or expansion tank fill cap. The engine is operated without the cap in this step.

Note : All coolant shutoff valves must be returned to their open positions once the system has been refilled and the deaeration process is ready to begin. Refer to the equipment manufacturer service information for specific instructions.

Wait 2 to 3 minutes without starting engine, to allow the system to deaerate and the fluid level to stabilize.

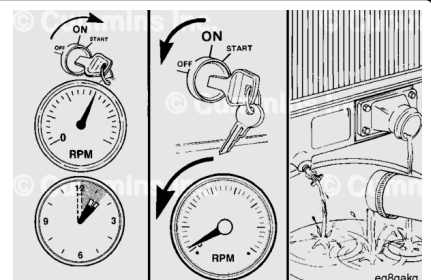
Add clean water to bring the level back to FULL COLD.

With the radiator cap or expansion tank fill cap off:

- Operate the engine at LOW IDLE for 2 minutes.
- Turn the engine OFF and add clean water to bring the level back to FULL COLD.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Install the radiator cap or expansion tank fill cap.

Operate the engine at HIGH IDLE and normal operating temperatures, at least 85°C [185°F], for 1 to 1½ hours.

Operate the engine at LOW IDLE for 2 minutes before shutting it down. This allows adequate cooling of pistons, cylinders, bearings, and turbocharger components.

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

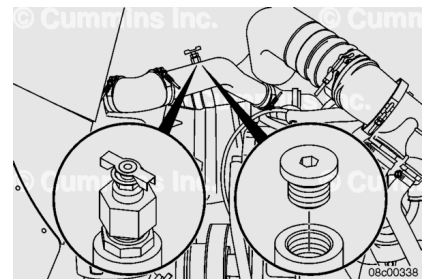
⚠ CAUTION ⚠

Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.

Note : Some applications can have a manual bleed valve or pipe plug that is required to be opened to properly fill the system. The upper radiator pipe is a common location.

Note : Some applications can have a pipe plug installed in place of the manual bleed valve.

If applicable, open the manual bleed valve or remove the pipe plug before filling the cooling system.

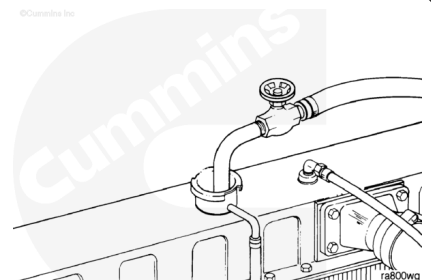


⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.

Note : Do not install the radiator cap or expansion tank fill cap. The engine is operated without the cap in this step.

Note : The system has a fill rate of 19 liters [5 gal] per minute.



Close the radiator drain valve(s).

Install the lower radiator hose(s). Refer to the equipment manufacturer service information.

Fill the cooling system with clean water.

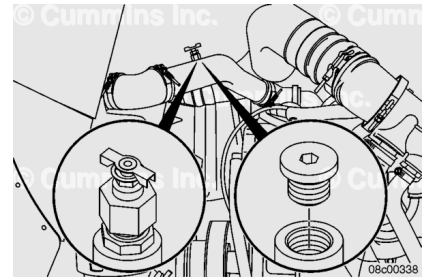
Wait 2 to 3 minutes without the starting engine to allow the system to deaerate and the fluid level to stabilize.

Add clean water to bring the level back to FULL COLD.

⚠ CAUTION ⚠

Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.

If applicable, install the manual bleed valve or the pipe plug after filling the cooling system. Refer to the equipment manufacturer service information.



⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.

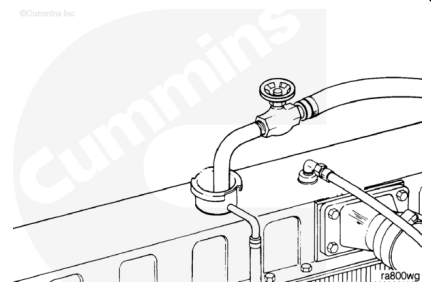
Note : Do not install the radiator cap or expansion tank fill cap. The engine is operated without the cap in this step.

Wait 2 to 3 minutes without starting the engine to allow the system to deaerate and the fluid level to stabilize.

Add clean water to bring the level back to FULL COLD.

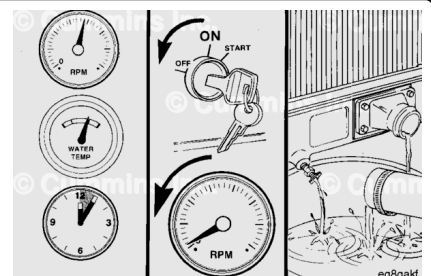
With the radiator or pressure cap off:

- Operate the engine at LOW IDLE for 2 minutes.
- Turn the engine OFF and add clean water to bring the level back to FULL COLD.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : All coolant shutoff valves should be returned to their open positions once the system has been refilled and the deaeration process is ready to begin. Refer to the equipment manufacturer service information for specific instructions.

Install the radiator cap or expansion tank fill cap.

Operate the engine at HIGH IDLE for 5 minutes with the coolant temperature above 85°C [185°F].

Operate the engine at LOW IDLE for 2 minutes before shutting it down. This allows adequate cooling of pistons, cylinders, bearings, and turbocharger components.

Shut the engine OFF. Allow it to cool to 50°C [120°F]. Drain the cooling system.

If the water being drained is still dirty, the system **must** be repeatedly flushed until the water is clean.

Fill

CM570

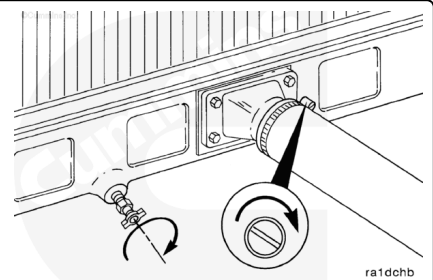
⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks, or serious engine damage can result.

Note : This method is to be used when the vehicle/equipment or engine is not equipped with a quick disconnect fitting in the cooling system package. If the engine or vehicle/equipment is equipped with a quick disconnect fitting, the Coolant Replacer Method must be used.

Close the radiator drain valve(s).

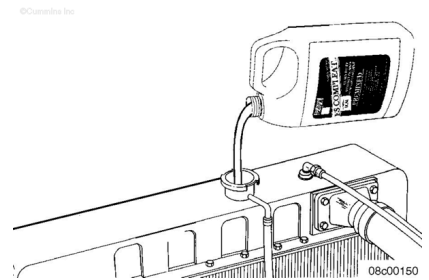
Install the lower radiator hose(s).



Tighten the hose clamps. Refer to the equipment manufacturer service information.

Cummins Inc. recommends the use of either a 50/50 mixture of good quality water and fully-formulated antifreeze, or fully-formulated coolant when filling the cooling system. The fully-formulated antifreeze or coolant **must** meet TMC RP329 or TMC RP330 specifications.

Good quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.



Water Quality

Calcium Magnesium (Hardness)	Maximum 170 ppm as (CaCO ₃ + MgCO ₃)
Chloride	40 ppm as (Cl)
Sulfate	100 ppm as (SO ₄)

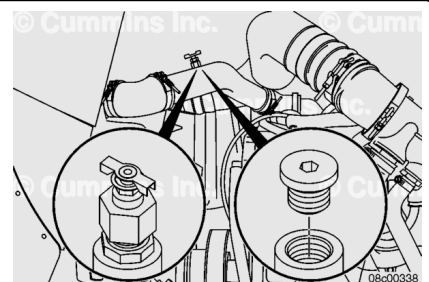
Cummins Inc. recommends the use of Fleetguard® ES COMPLEAT™. It is available in glycol forms (ethylene and propylene) and complies with TMC RP329 and RP330 standards.

Refer to Cummins® Coolant Requirement and Maintenance, Bulletin 3666132, ([/qs3/pubsys2/xml/en/bulletin/3666132.html](http://qs3/pubsys2/xml/en/bulletin/3666132.html)) for more engine coolant specifications.



⚠ CAUTION ⚠

Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



Note : Some applications can have a manual bleed valve or pipe plug that is required to be opened to properly fill the system. The upper radiator pipe is a common location.

Note : Some applications can have a pipe plug installed in place of the manual bleed valve.

If applicable, open the manual bleed valve or remove the pipe plug before filling the cooling system.

Note : See equipment manufacturer service information for special coolant fill requirements. Special instructions may also be located near the cooling system access point or fill door on the vehicle.

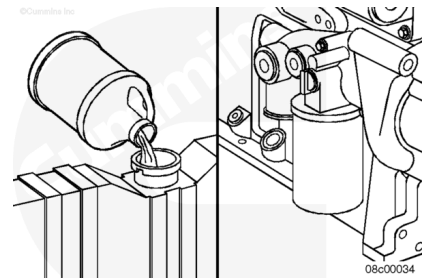
Note : The system has a designed fill rate of 19 liters [5 gal] per minute.

Note : Not all engines are assembled with a coolant filter. The use of coolant filter is optional when using coolant meeting Cummins® coolant requirements.

Fill the cooling system with heavy-duty coolant and install the correct service filter.

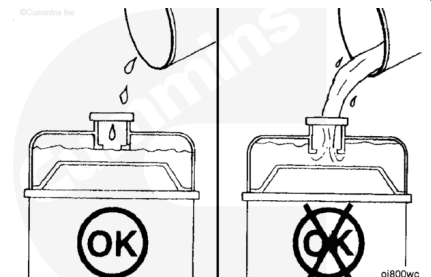
Identify the correct filter. For automotive applications, use the Signature™ and ISX Operation and Maintenance Manual, Bulletin 3666251. Refer to Procedure 018-004 in Section V. (</qs3/pubsys2/xml/en/procedures/00/00-018-004.html>)

For industrial applications, use the Industrial and Power Generation QSX15 Engines Operation and Maintenance Manual, Bulletin 3666423. Refer to Procedure 018-004 in Section V. (</qs3/pubsys2/xml/en/procedures/101/101-018-004.html>)



Note : Some radiators have two fill locations, both of which must be filled when the cooling system has been drained.

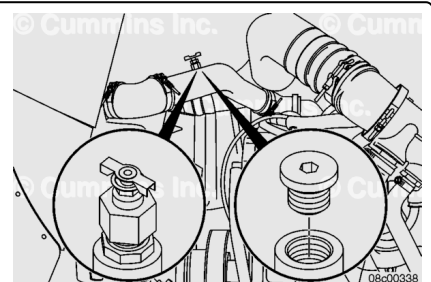
Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.



⚠ CAUTION ⚠

Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.

If applicable, install the manual bleed valve or the pipe plug after filling the cooling system. Refer to the equipment manufacturer service information.



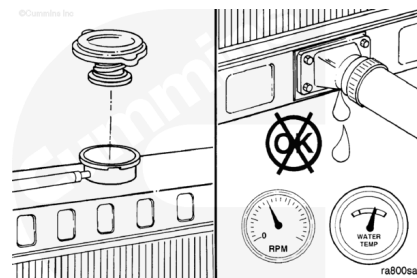
Note : All coolant shutoff valves should be returned to their open positions once the system has been refilled and the deaeration process is ready to begin. Refer to the equipment manufacturer service information for specific instructions.

Install the radiator cap or expansion tank fill cap.

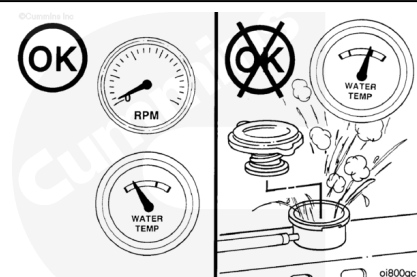
Turn the cab heater temperature switch to high to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

Operate the engine at 1600 rpm for 10 minutes.

Operate the engine at LOW IDLE for 2 minutes before shutting it down. This allows adequate cool down of pistons, cylinders, bearings, and turbocharger components.

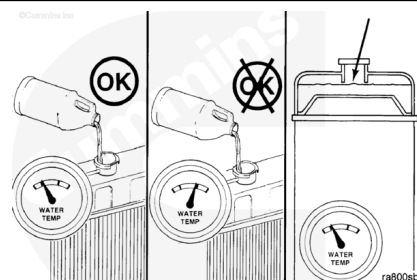


Shut the engine OFF and allow it to cool.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ CAUTION ⚠

Do not add cold coolant to a hot engine. This can cause engine casting damage. Allow the engine to cool to below 50°C [120°F] before adding coolant.

Note : If all coolant drained from the system was collected, the same volume or more **must** go back into the system. If any drained coolant remains after filling, this is an indication of an air pocket which must be purged before returning the vehicle to service.

Remove the radiator or expansion tank fill cap and check the coolant level according to the equipment manufacturer service information.

Add coolant if necessary. Do **not** overfill.

Automotive with CM870 and CM871

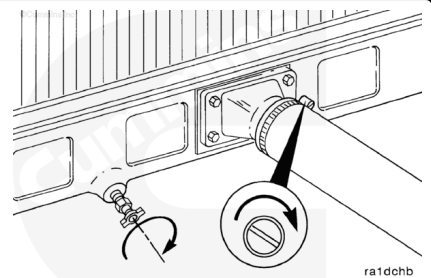
⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.

Close the radiator drain valve(s).

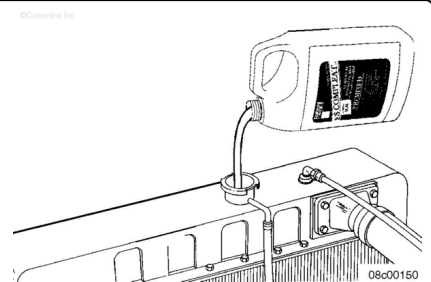
Install the lower radiator hose(s).

Tighten the hose clamps. Refer to the equipment manufacturer service information.



Cummins Inc. recommends the use of either a 50/50 mixture of good-quality water and fully-formulated antifreeze, or fully-formulated coolant when filling the cooling system. The fully-formulated antifreeze or coolant **must** meet TMC RP329 or TMC RP330 specifications.

Good-quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.



Water Quality

Calcium Magnesium (Hardness)	Maximum 170 ppm as (CaCO ₃ + MgCO ₃)
Chloride	40 ppm as (Cl)
Sulfate	100 ppm as (SO ₄)

Cummins Inc. recommends the use of Fleetguard® ES COMPLEAT™. It is available in glycol forms (ethylene and propylene) and complies with TMC RP329 and RP330 standards.

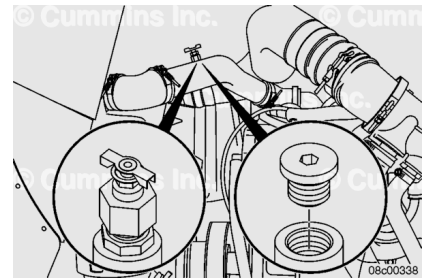
Refer to Cummins® Coolant Requirement and Maintenance, Bulletin 3666132, ([/qs3/pubsys2/xml/en/bulletin/3666132.html](http://qs3/pubsys2/xml/en/bulletin/3666132.html)) for more



engine coolant specifications.

⚠ CAUTION ⚠

Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



Note : Some applications can have a manual bleed valve or pipe plug that is required to be opened to properly fill the system. The upper radiator pipe is a common location.

Note : Some applications can have a pipe plug installed in place of the manual bleed valve.

If applicable, open the manual bleed valve or remove the pipe plug before filling the cooling system.

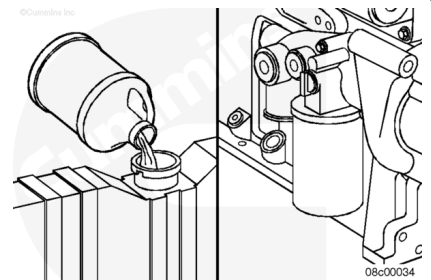
Note : Refer to the OEM service manual for special coolant fill requirements. Special instructions may also be located near the cooling system access point or fill door on the vehicle.

Note : The system has a fill rate of 19 liters [5 gal] per minute.

Fill the cooling system with heavy-duty coolant and install the correct service filter.

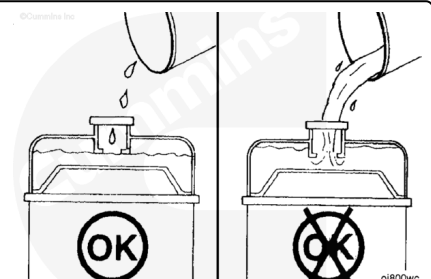
Identify the correct filter. For automotive applications, use the Signature™ and ISX Operation and Maintenance Manual, Bulletin 3666251. Refer to Procedure 018-004 in Section V. (</qs3/pubsys2/xml/en/procedures/00/00-018-004.html>)

For industrial applications, use the Industrial and Power Generation QSX15 Engine Operation and Maintenance Manual, Bulletin 3666423. Refer to Procedure 018-004 in Section V. (</qs3/pubsys2/xml/en/procedures/101/101-018-004.html>)



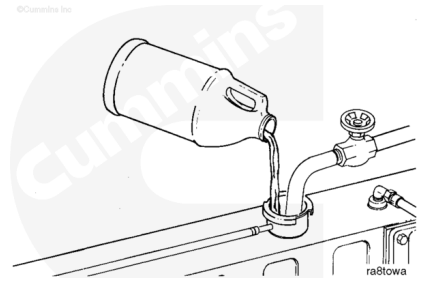
Note : Some radiators have two fill locations, both of which must be filled when the cooling system has been drained.

Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.



⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.



Note : Do not install the radiator cap or expansion tank fill cap. The engine is operated without the cap in this step.

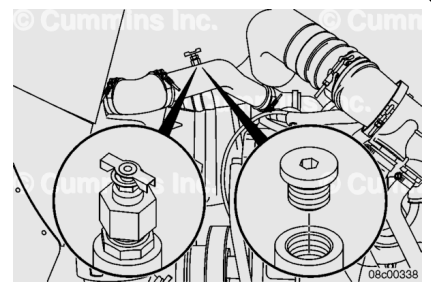
Turn the cab heater temperature switch to high to allow maximum coolant flow through the heater core. The blower does not have to be on.

Wait 2 to 3 minutes without starting engine to allow the system to deaerate and the fluid level to stabilize.

Add coolant to bring the level back to FULL COLD.

⚠ CAUTION ⚠

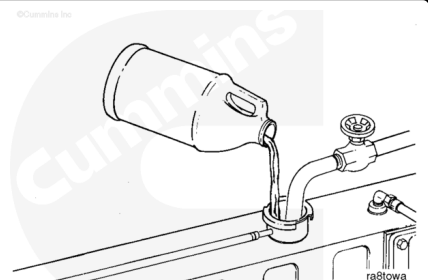
Opening the manual bleed valve or removing the pipe plug on applicable installations is critical. Failure to do so can result in engine damage.



If applicable, install the manual bleed valve or the pipe plug after filling the cooling system. Refer to the equipment manufacturer service information.

⚠ CAUTION ⚠

The cooling system must be filled properly to prevent air locks or serious engine damage can result.



Note : Do not install the radiator cap or expansion tank fill cap. The engine is operated without the cap in this step.

With the radiator cap or expansion tank fill cap off:

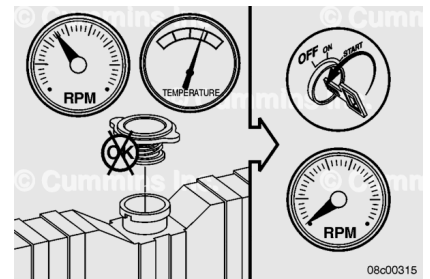
- Operate the engine at LOW IDLE for 2 minutes.
- Turn the engine OFF and add clean water to bring the level back to FULL COLD.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Note : All coolant shutoff valves should be returned to their open positions once the system has been refilled and the deaeration process is ready to begin. Refer to the equipment manufacturer service information for specific instructions.

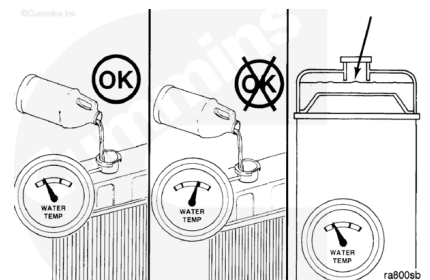
Operate the engine at HIGH IDLE until the thermostat opens, at least 93°C [200°F].

Operate the engine at LOW IDLE for 2 minutes before shutting it down. This allows adequate cool down of pistons, cylinders, bearings, and turbocharger components.

Shut the engine OFF and allow to cool.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ CAUTION ⚠

Do not add cold coolant to a hot engine. This can cause engine casting damage. Allow the engine to cool to below 50°C [120°F] before adding coolant.

Note : If all coolant drained from the system was collected, the same volume or more must go back into the system. If any drained coolant remains after filling, this is an indication of an air pocket which must be purged before returning the vehicle to service.

Remove the radiator cap or expansion tank fill cap and check the coolant level according to the equipment manufacturer service information.

Add coolant if necessary. Do **not** overfill.

See Figure 1, illustrating the steps of the coolant fill and deaeration process.

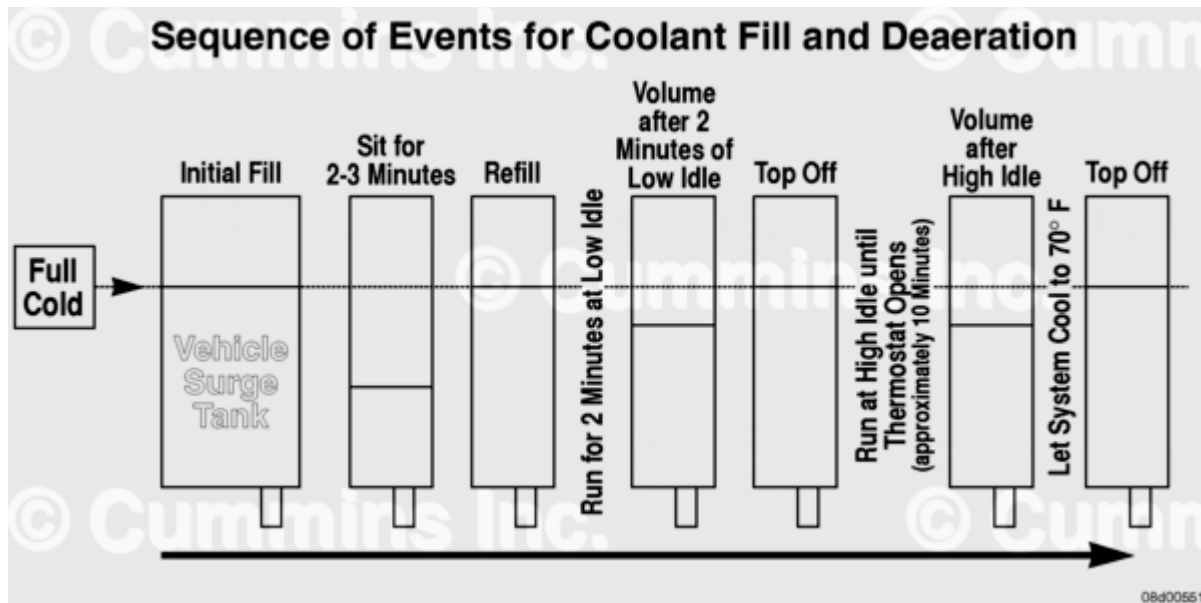


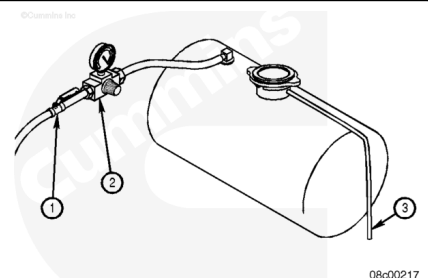
Figure 1: Coolant Fill and Deaeration Process

Pressure Test

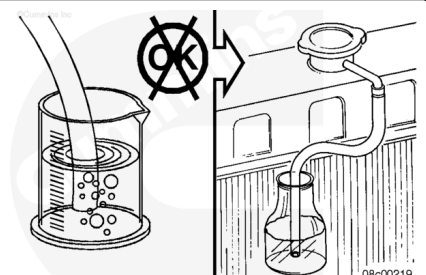
Connect the air line in place of a vent line on the surge tank.

Install an air regulator (2) between the ball valve (1) of the air line and the surge tank connection.

Attach a hose to the radiator overflow connection (3).

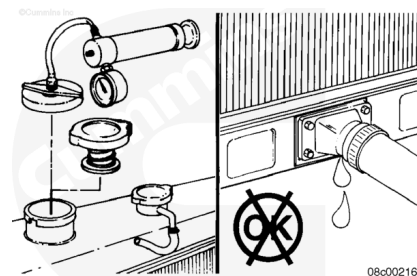


Place the free end of the hose below the water level in a container of water.



⚠ CAUTION ⚠

Do not apply more than 140 kPa [20 psi] air pressure to the cooling system. The water pump seal can be damaged.



Pressurize the cooling system slowly to the rated radiator cap pressure until bubbles can be seen in the overflow bottle.

If the bubbles start at a lower pressure than the minimum recommended cap pressure (See Cooling System Specifications), inspect the radiator pressure cap. Refer to Procedure 008-047 in Section 8.

(</qs3/pubsys2/xml/en/procedures/10/10-008-047.html>)

Replace the radiator pressure cap, if necessary.

